**13** Solid **A** and solid **B** are similar.

The ratio of the volume of solid **A** to the volume of solid **B** is 27 : 1000

The surface area of solid **A** is 810 cm$^2$

Calculate the surface area of solid **B**.

Volume ratio A : B = 27 : 1000 = $3^3$ : $10^3$

Length ratio A : B = 3 : 10

**1 mark**
Cube roots for lengths.

Area ratio A : B = $3^2$ : $10^2$ = 9 : 100

**1 mark**
Square for areas.

Surface area B = 810 ÷ 9 × 100 = 9000 cm$^2$

**1 mark**
9000 cm$^2$.

..... cm$^2$

**(Total for Question 13 is 3 marks)**

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